

Random Signal Theory–EEE 554

Lecture Notes 3

Joint Moments

[Reading: Kay, Sections 7.7, 7.8, 7.10, 12.7, 12.8, 12.10]

Expected Value of Function of Joint Random Variables

To find the mean of Z , where $Z = g(X, Y)$,

$$E[Z] = E[g(X, Y)] = \int_{S_Z} z f_Z(z) \, dz = \int \int_{S_{X,Y}} g(x, y) f_{XY}(x, y) \, dx \, dy$$

Example: Let a $Z = aX + bY$, find mean of Z

Using linearity and marginal PDFs,

$$\begin{aligned} \mu_z = E[Z] &= E[aX + bY] = \int \int (ax + by) f_{XY}(x, y) \, dx \, dy \\ &= a \int \int x f_{XY}(x, y) \, dx \, dy + b \int \int y f_{XY}(x, y) \, dx \, dy \\ &= a \int x \left(\int f_{XY}(x, y) \, dy \right) dx + b \int y \left(\int f_{XY}(x, y) \, dx \right) dy \\ &= a \int x f_X(x) \, dx + b \int y f_Y(y) \, dy = a E[X] + b E[Y] = a \mu_x + b \mu_y \end{aligned}$$

Example:

$Z = g(X, Y) = XY$, when X and Y are not independent

$$\begin{aligned}\mu_z = E[Z] &= \int \int xy f_{XY}(x, y) \, dx \, dy \\ &\neq E[X] E[Y]\end{aligned}$$

Example:

$Z = g(X, Y) = XY$, when X and Y are independent

$$\begin{aligned}\mu_z = E[Z] &= \int \int xy f_{XY}(x, y) \, dx \, dy \\ &= \int \int xy f_X(x) f_Y(y) \, dx \, dy \\ &= \left(\int x f_X(x) \, dx \right) \left(\int y f_Y(y) \, dy \right) \\ &= E[X] E[Y] = \mu_x \mu_y\end{aligned}$$

Example: Find $\text{Var}(Z)$ when $Z = g(X, Y) = X + Y$

Since $\text{Var}(Z) = E[Z^2] - E^2[Z] = \text{Var}(X + Y)$, then

$$\begin{aligned}\text{Var}(X + Y) &= E[(X + Y)^2] - E^2[X + Y] = E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2 \\&= E[X^2] + 2E[XY] + E[Y^2] - (E^2[X] + 2E[X]E[Y] + E^2[Y]) \\&= (E[X^2] - E^2[X]) + (E[Y^2] - E^2[Y]) + 2(E[XY] - E[X]E[Y]) \\&= \text{Var}(X) + \text{Var}(Y) + 2C_{XY}\end{aligned}$$

where covariance $C_{XY} = E[XY] - E[X]E[Y]$ is the covariance between X and Y

If X and Y independent, it follows that $E[XY] = E[X]E[Y]$

$\Rightarrow C_{XY} = E[XY] - E[X]E[Y] = 0$ and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Joint Statistical Measures

- Covariance: first joint central moment of X and Y

$$\begin{aligned}C_{XY} &= E[(X - \mu_x)(Y - \mu_y)] \\&= E[XY - \mu_y X - \mu_x Y + \mu_x \mu_y] = E[XY] - \mu_y E[X] - \mu_x E[Y] + \mu_x \mu_y \\&= E[XY] - 2\mu_x \mu_y + \mu_x \mu_y = \boxed{E[XY] - \mu_x \mu_y}\end{aligned}$$

- Correlation: obtained as $E[XY]$

Correlation $\neq$ covariance unless $\mu_x = \mu_y = 0$

- Correlation coefficient: extent by which Y is predicted as a linear combination of X

$$\rho_{XY} = \frac{C_{XY}}{\sigma_x \sigma_y} = \frac{E[XY] - \mu_x \mu_y}{\sigma_x \sigma_y}$$

where it can be shown that $-1 \leq \rho_{XY} \leq 1$ (equivalently, $-\sigma_x \sigma_y \leq C_{XY} \leq \sigma_x \sigma_y$)

Properties

- ▶ If X & Y are uncorrelated $\Rightarrow \rho_{XY} = 0 \Rightarrow C_{XY} = 0 \Rightarrow E[XY] = \mu_x \mu_y$
- ▶ If X & Y are independent $\Rightarrow E[XY] = \mu_x \mu_y \Rightarrow C_{XY} = 0$
 $\Rightarrow X$ and Y are uncorrelated
- ▶ If X & Y are independent, then they are also uncorrelated
- ▶ If X & Y are uncorrelated, not necessarily independent
for independence, must show that $f_{XY}(x, y) = f_X(x) f_Y(y)$
- ▶ X & Y are orthogonal if $E[XY] = 0$

Special case:

if X & Y are uncorrelated and $\mu_x = \mu_y = 0$ then $\Rightarrow C_{XY} = 0$
 $\Rightarrow E[XY] - \mu_x \mu_y = 0 \quad E[XY] = \mu_x \mu_y = 0, \Rightarrow X$ and Y are orthogonal

Example:

Find correlation $E[XY]$ and covariance C_{XY} when $f(x, y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & \text{otherwise} \end{cases}$

Correlation:

$$\begin{aligned} E[XY] &= \int_{x=0}^{\infty} \int_{y=0}^x xy f(x, y) dy dx = \int_{x=0}^{\infty} \int_{y=0}^x xy 2e^{-x}e^{-y} dy dx \\ &= \int_{x=0}^{\infty} 2xe^{-x} \left(\int_{y=0}^x ye^{-y} dy \right) dx = \int_{x=0}^{\infty} 2xe^{-x} (1 - xe^{-x} - e^{-x}) dx \\ &= \int_{x=0}^{\infty} (2xe^{-x} - 2x^2e^{-2x} - 2xe^{-2x}) dx = 2 - (4/2^3) - 2/(2^2) = 1 \end{aligned}$$

Above used

$$\int x e^{kx} dx = \frac{(kx - 1)}{k^2} e^{kx} \quad \text{and} \quad \int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}} \text{ for } n > 0$$

Covariance:

$$C_{XY} = E[XY] - \mu_x \mu_y = 1 - (1.5)(0.5) = 0.25$$

since means μ_x and μ_y are computed as

$$\mu_x = \int_0^{\infty} x f_X(x) dx = \int_0^{\infty} x 2 e^{-x} (1 - e^{-x}) dx = 1.5$$

$$\mu_y = \int_0^{\infty} y f_Y(y) dy = \int_0^{\infty} y 2 e^{-2y} dy = 0.5$$

using marginal PDFs (derivation in Notes 3)

$$f_X(x) = 2e^{-x}(1 - e^{-x}) u(x) \quad \text{and} \quad f_Y(y) = 2e^{-2y} u(y)$$

Joint moments

For jointly distributed RVs, k, ℓ th joint moment

$$E[X^k Y^\ell] = \int \int_{S_{X,Y}} x^k y^\ell f_{XY}(x, y) \, dx \, dy$$

and k, ℓ th joint central moment

$$E[(X - \mu_X)^k (Y - \mu_Y)^\ell] = \int \int_{S_{X,Y}} (x - \mu_X)^k (y - \mu_Y)^\ell f_{XY}(x, y) \, dx \, dy$$

Joint Characteristic Function

- ▶ Two-dimensional Fourier transform (FT) of joint PDF

$$\Phi_{X,Y}(\omega_1, \omega_2) = E\left[e^{j\omega_1 X} e^{j\omega_2 Y}\right] = \int \int_{S_{X,Y}} e^{j\omega_1 x} e^{j\omega_2 y} f_{XY}(x, y) \, dx \, dy$$

- ▶ Marginal Characteristic Functions

$$\Phi_X(\omega) = \int_{S_X} e^{j\omega x} f_X(x) \, dx = \Phi_{X,Y}(\omega, 0)$$

$$\Phi_Y(\omega) = \int_{S_Y} e^{j\omega y} f_Y(y) \, dy = \Phi_{X,Y}(0, \omega)$$

- ▶ If X and Y independent, $f_{XY}(x, y) = f_X(x) f_Y(y)$ and $\Phi_{XY}(\omega_1, \omega_2) = \Phi_X(\omega_1) \Phi_Y(\omega_2)$

- ▶ If X and Y independent and $Z = X + Y$, and using FT properties

$$f_Z(z) = f_X(z) \star f_Y(z) \text{ (convolution) and } \Phi_Z(\omega) = \Phi_X(\omega) \Phi_Y(\omega) \text{ (multiplication)}$$